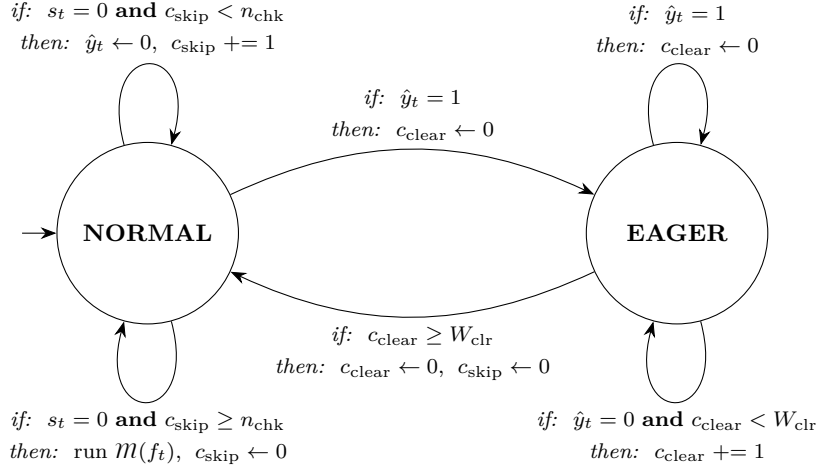


Skip Module Hyperparameter Selection

1 Algorithm

tikz



Symbol	Description
$s_t \in \{0, 1\}$	Skip decision from $\mathcal{S}$: $s_t=0$ no motion detected (skip inference); $s_t=1$ motion detected (run inference)
$\hat{y}_t \in \{0, 1\}$	Prediction from $\mathcal{M}$: $\hat{y}_t=1$ fire/smoke; $\hat{y}_t=0$ negative
c_{skip}	Number of consecutively skipped frames (reset to 0 on any inference call)
c_{clear}	Number of consecutive negative predictions accumulated in EAGER mode
n_{chk}	Maximum consecutive skips allowed before a forced inference check is issued
W_{clr}	Number of consecutive negative predictions required to exit EAGER mode

Figure 1: State machine of the eager mode extension. Each transition is annotated with a *guard* (condition to fire) and an *action* (variable updates on transition). In NORMAL mode, $\mathcal{S}$ gates inference: no-motion frames are skipped ($c_{\text{skip}} += 1$) until $c_{\text{skip}} \geq n_{\text{chk}}$, at which point a forced inference check resets c_{skip} . Any fire or smoke prediction immediately enters EAGER mode, bypassing $\mathcal{S}$ and running $\mathcal{M}$ on every frame. EAGER exits to NORMAL only after W_{clr} consecutive negative predictions confirm the scene is safe.